SPE-229862-MS

A Technique for Assessing Lithium Concentration in Saline Porous Media Using Pulsed Neutron Logging

Y. Kim, Baker Hughes, Dhahran, Kingdom of Saudi Arabia; F. Inanc, Baker Hughes, Houston, United States of America

 DOI 10.2118/229862-MS

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

The significance of lithium extraction is increasingly recognized due to the current and projected demand across various industrial sectors. However, the technologies for the identification and quantification of lithium concentration within saline porous media remain in their nascent stages and present considerable challenges. We have developed a method for assessing lithium concentration via in-well pulsed neutron logging.

This method delineates the most efficient target zone for lithium extraction on a continuous well logging basis, whether in openhole or cased wellbores. The thermal neutron-alpha (n,α) reaction cross section (i.e., Sigma) of lithium-6 is as high as approximately 940 barns. However, when a thermal neutron is absorbed by lithium, no detectable characteristic gamma-ray emission occurs, which can be quantified via pulsed neutron logging instruments. In contrast, the high Sigma value of lithium substantially influences neutron lifetime. Nevertheless, it suffers from uniqueness because other elements, such as chlorine and boron—commonly found concomitantly with lithium—that also significantly affect neutron lifetime. Therefore, it is essential to measure proxy elements associated with lithium presence that can be detected through pulsed neutron well logging and to account for their contributions to isolate lithium signals within the time spectra.

This method involves integrating energy and time spectra. Initially, energy spectra are employed to determine the concentrations of chlorine and boron, utilizing spectrum deconvolution techniques. Subsequently, two distinct approaches are applied for lithium concentration estimation: (1) a process to remove the influences of chlorine and boron from the time spectra, thereby isolating the lithium signal—referred to as time spectrum stripping; (2) forward modeling of pulsed neutron measurements extracted from the time spectra using the Monte Carlo N-Particle (MCNP) transport code to predict the relationship among lithium concentration, boron concentration, and porosity. The Total-to-Capture ratio is developed as an effective pulsed neutron measurement, leveraging fast, epithermal, and thermal neutrons for gamma-ray production. Ultimately, this method facilitates the derivation of lithium concentration as a continuous well log. In situ sampling of lithium-dissolved brine, aimed at estimating lithium concentration, serves as a valuable complementary technique for validating the lithium logs obtained through pulsed neutron measurements.

The benefits of employing pulsed neutron logging technology include (1) the cost-effectiveness of data acquisition and (2) the enhanced accessibility of information pertaining to lithium on a continuous depth basis throughout the targeted formation, as opposed to the discrete depths encountered in fluid sampling or coring techniques.

Furthermore, pulsed neutron well logging functions as a non-destructive surveillance method for casing and cement prior to final decision-making on perforation interval selection. The lithium concentration log enables operators to identify the most productive zones for lithium extraction.

Introduction

This paper discusses nuclear measurement techniques for determining lithium concentrations in brines during drilling and production in oil fields. The initiative to develop this method stems from the increasing demand for lithium in various industries, most notably in batteries for next-generation hybrid and electric vehicles, as well as renewable energy storage systems. Although lithium exists in various forms, extracting it from brines with concentrations exceeding 200 parts per million (ppm) is generally considered economically feasible. The low concentration of lithium poses a significant challenge in accurately evaluating its levels. In situ fluid sampling to assess lithium concentration is a viable option; however, developing sensors is time-consuming and costly, and sampling and analysis are performed on a station basis. Similarly, core analysis provides discrete depth-based data collection, which is a costly approach. Pulsed neutron well logging emerges as a compelling alternative solution. It provides continuous assessment profiles of geological formations, and its measurement approach is non-destructive in both open-hole and cased-hole completion wells. Furthermore, pulsed neutron logging enables rapid measurements, which are particularly advantageous in exploration scenarios where efficiency is a key element.

Fundamentals of Pulsed Neutron Well Logging

This section provides a brief overview of the fundamentals of pulsed neutron well logging in oil and gas fields. This includes the main components of the pulsed neutron tool, multifunctional data acquisition, well logging speeds, and applications.

Pulsed Neutron Well Logging Tool and Data Acquisition

The pulsed neutron generator was first developed in 1954 (Youmans et al., 1959), and the first pulsed neutron well logging instrument was introduced in the early 1960s (Youmans et al., 1964; Randall et al., 1985). The outer diameter of the first commercial pulsed neutron well logging instrument was 3-5/8 inches. The outer diameter of the pulsed neutron tool has been redesigned to be slimmer in subsequent development, facilitating through-tubing deployment.

The essential components of a pulsed neutron tool comprise a pulsed neutron generator (PNG), scintillator detectors, and associated electronics. The PNG employs a deuterium-tritium fusion reaction to produce high-energy neutrons, specifically fast neutrons with an energy of 14.2 (mega-electron volts (MeV). Gamma rays are generated as a result of interactions between neutrons and elements within the measuring medium. These gamma rays are subsequently registered using pulsed neutron detectors and electronic systems in relation to the PNG bursts or their respective energy levels. The first pulsed neutron tool had a single detector; however, the industry's first three-detector pulsed neutron tool (i.e., short-, long-, and extra-long-spaced detectors) was developed in 1999 (Gilchrist et al., 1999). The extra-long-spaced detector is placed farther from the PNG than the long-spaced detector, which increases the spatial measurement volumes and thereby enhances formation sensitivity, especially in gas-filled formations.

The time spectrum at each of the three detectors is constructed using the gamma rays measured over time, and analyzing the temporal decay spectra is instrumental in determining the thermal neutron capture cross section. Conversely, energy spectra are based on the energies of the induced gamma rays.

Pulsed Neutron Well Logging Speed

The well logging speed is dependent on the primary acquisition of spectra (i.e., time or energy spectra) and specifications of pulsed neutron tools. Pulsed neutron capture (PNC) logging primarily acquires time spectral data, whereas carbon/oxygen (C/O) logging records energy spectra. The logging speed of PNC logging exceeds that of C/O logging. Typically, PNC logging is conducted at a rate of 10–30 feet per minute (ft/min), whereas C/O logging is commonly logged at 2–6 ft/min.

Logging speed is also influenced by the output of the pulsed neutron source, the performance of scintillation detector crystals, and the efficiency of electronics used for gamma-ray processing. The first slimhole multidetector pulsed neutron tool has a sodium iodide (NaI) scintillation detector crystal and analog electronics. However, the next-generation pulsed neutron tool has been enhanced with a PNG that produces twice the number of fast neutrons compared to the previous PNG, a denser lanthanum bromide ($LaBr_3$) detector crystal, and digital electronics (Nardiello et al., 2022; Kim et al., 2023a; Akagbosu et al., 2023; McGlynn et al., 2023).

This advancement in technology enables faster data acquisition. The previous-generation pulsed neutron tool can log PNC datasets at 10–20 ft/min, while the next-generation pulsed neutron tool can log data of comparable quality at 20–30 ft/min. Likewise, the same-quality C/O dataset can be recorded at 2 ft/min with the legacy tool and at 6 ft/min with the next-generation tool. The enhancement of logging speed was accomplished not only for the existing acquisition modes, such as PNC and C/O modes, but also by developing a new acquisition mode that facilitates the simultaneous collection of energy and time spectra (i.e., a simultaneous mode). Historically, acquiring high-quality energy spectra and time spectra required two separate logging runs; however, the three improvements in the next-generation pulsed neutron tool enabled recording both types of spectral data in a single run without compromising data quality at the same logging speed as C/O mode (i.e., typically, acquiring a single pass at 2 ft/min or three passes at 6 ft/min).

Applications of Pulsed Neutron Well Logs

Reservoir Surveillance and Monitoring. Pulsed neutron logging has been widely adopted to surveil the reservoir behind casing, yet it is also applicable for barefoot wells. Reservoir surveillance and monitoring are the main applications of pulsed neutron well logging. Sigma log analysis is used for water saturation calculation if the water salinity is high and known. When water salinity is low, mixed, or unknown, such as in the post-waterflooding stage, C/O logging is required (Kim et al., 2023c). Sigma-based saturation analysis was first used in 1963, and C/O-based saturation analysis has evolved since 1973.

For gas reservoir surveillance and monitoring, Sigma log can also provide gas saturation analysis in a saline water environment. If the water salinity is low, the traditional pulsed neutron log overlay technique has been used. Later, with the advent of the multidetector pulsed neutron tool, a salinity-insensitive yet hydrogen-sensitive ratio measurement has been introduced as a viable solution for formation gas volume quantification (Trcka et al., 2006; Inanc et al., 2009; Kim et al., 2023b). Expanding the interpretation algorithm of gas quantification enables the estimation of reservoir pressure depletion without perforating the target reservoirs (Nardiello et al., 2017; Kim et al., 2023d; 2025b).

Three-phase reservoir fluid saturation analysis is also applicable using pulsed neutron logs. The analysis techniques include both sequential and simultaneous approaches (Kim & Chace, 2016). Reservoir monitoring for CO_2-enhanced oil recovery (CO_2-EOR) (Conner et al., 2018; Kim et al., 2024b), water-alternating-gas/CO_2 (WAG) or gas cap evaluation in mature oil reservoirs employs a three-phase reservoir saturation analysis method (Kim et al., 2017; 2024a).

Furthermore, monitoring projects such as breakthrough of freshwater or CO_2, CO_2 plume profiling, CO_2 storage monitoring (Kim et al., 2025a), and steam chamber growth (Kim et al., 2021; 2023a), among others, utilize pulsed neutron logging technology.

Formation Evaluation. Evaluation of formation petrophysical properties other than formation saturation can be performed using pulsed neutron logs. A pulsed neutron porosity log extract from the time spectra allows for the evaluation of formation porosity variation when compared to previous neutron porosity logs, especially the open-hole neutron porosity log.

Silicon activation logging provides the volume of silicon. This method can be employed as a general lithology indicator in conjunction with other pulsed neutron measurements, such as Sigma (i.e., differentiation of shale, sand, and carbonate). Furthermore, the silicon activation log can be used to assess gravel pack completion and analyze materials in the annular space.

Pulsed Neutron Production Logging. When centerline and array production logging (PL) sensors are unsuitable for capturing multiphase flow properties owing to challenging borehole conditions—such as the accumulation of debris, asphaltene, and other well materials—pulsed neutron well logging can serve as an effective alternative solution. Pulsed neutron logging is not limited by the well materials, which can cause sensor clogging and malfunction; thus, it can provide measurements to characterize fractions of multi-fluid components (i.e., a three-phase holdup) in the borehole and water velocity (Chace et al., 1994; Trcka et al., 1996; Kim et al., 2022).

Lithium Identification Principles and Pulsed Neutron Technique

We propose a hybrid approach as the crux of lithium identification. This approach utilizes energy spectra first and then transitions to time spectra.

Lithium-Influencing Neutron Behavior and Time Spectra Approach

Nuclear methods for estimating the concentration of an element in a sample typically rely on measurements obtained from energy spectra. Specific nuclides, when exposed to neutrons, emit characteristic gamma rays, which facilitate the identification and quantification of that radioisotope within the sample. However, not all elements contain nuclides suitable for energy spectra measurements.

Although it is feasible to estimate lithium concentration through nuclear reactions, this approach presents particular challenges. One such challenge is that lithium does not emit neutron-induced gamma rays that are useful for characterizing lithium within the sample. Since lithium lacks nuclides that emit characteristic gamma rays, this limitation affects the use of energy spectra. Therefore, the measurement approach must depend on time spectra.

Lithium substantially influences the neutron time behavior. Natural lithium contains approximately 7.5% of Lithium-6 (^{6}Li), which has a thermal neutron (n,α) cross section of roughly 941 barns. Consequently, ^{6}Li exerts a significant impact on the neutron time behavior of the time spectra. Although energy spectrum-based techniques are inadequate for predicting lithium concentration, the influence of ^{6}Li on neutron time behavior provides an indirect method for estimating lithium levels.

A Hybrid Method—Integrating Energy and Time Spectra

Another significant challenge in predicting lithium concentration is that lithium-rich brines also contain boron. Boron-10 (^{10}B), which accounts for 19.9% of boron, has a thermal (n,α) cross section of approximately 3,838 barns. Chlorine-35 (^{35}Cl), comprising 75.77% of chlorine, has a cross section of 43.6 barns; it remains substantially lower than that of ^{10}B and ^{6}Li. However, the thermal neutron cross section for ^{35}Cl is relatively high when compared to those of formation matrix elements such as silicon-28 (169 millibarns), calcium-44 (880 millibarns), magnesium-24 (54 millibarns), or oxygen (0.19 millibarns). Both ^{10}B and ^{35}Cl considerably influence the time spectra, akin to lithium. Consequently, reliance solely on time spectrum measurements is inadequate for the precise prediction of lithium concentrations.

An alternative approach to using only the time spectrum or the energy spectrum for predicting lithium concentration is to implement a hybrid method. In this approach, energy spectrum data are used to estimate

^{10}B and ^{35}Cl concentrations, which are then employed to assess their respective time behavior parameters. This allows for the isolation of time behavior parameters attributed to lithium once the parameters of ^{10}B and ^{35}Cl are known. The isolated lithium time behavior can subsequently be used to predict lithium concentration in the brine.

Pulsed Neutron Logging Data-Based Hybrid Method for Lithium Concentration Delineation

Energy Spectrum

The first step in measurement is obtaining an energy spectrum from a target formation using the **next-generation pulsed neutron well logging tool**. A **simultaneous acquisition mode** can be employed, obviating the necessity for an additional logging operation required to obtain the time spectrum essential for **subsequent analysis**. A typical spectrum from a brine-filled sandstone formation exposed to a neutron beam (i.e., in this case, a pulsed neutron source) that includes fast, epithermal, and thermal neutrons is shown in Fig. 1. Each of the three detectors in the pulsed neutron tool provides an energy spectrum. The spectrum exhibiting the highest number of gamma-ray counts corresponds to the sequence of short-, long-, and extra-long-spaced detectors, following the order of increasing proximity to the PNG. In this study, we use the spectrum obtained from the short-spaced detector. Note that a pulsed neutron tool can generate a purely thermal neutron beam; other sources, such as chemical neutron sources, invariably emit a mixture of fast, epithermal, and thermal neutrons. This mixture also occurs during the burst gate of the pulsed neutron tool's pulsing scheme. The resultant gamma spectrum will show different peaks depending on the neutron source.

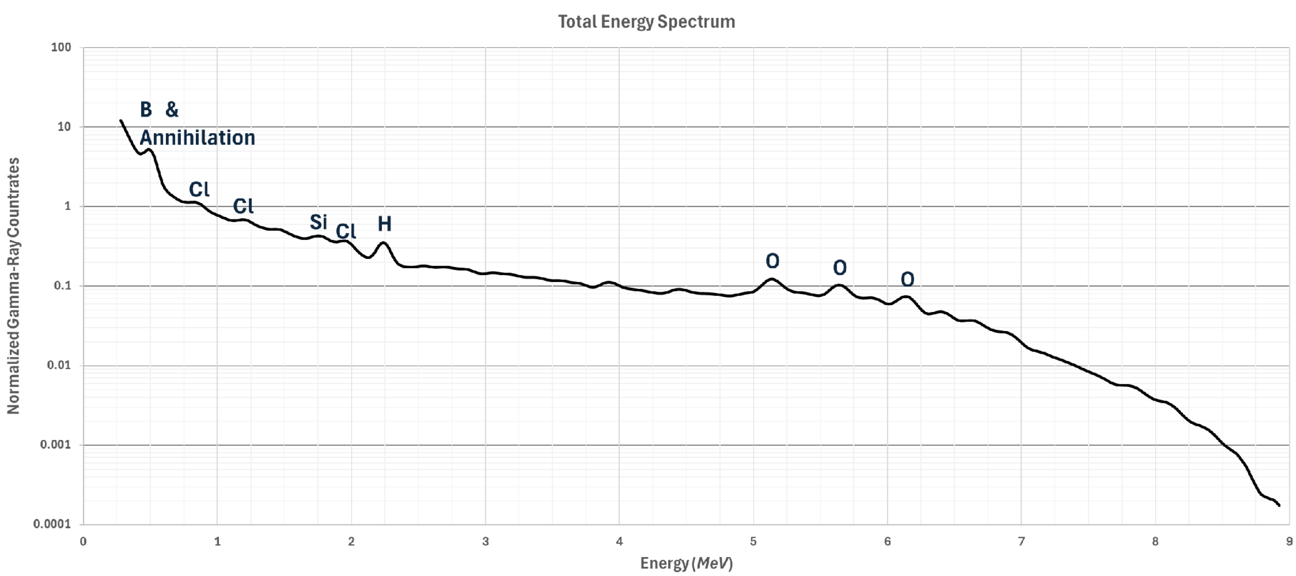

Figure 1—Neutron-induced gamma-ray energy spectrum from a brine-filled sandstone.

This energy spectrum was simulated using the Monte Carlo N-Particle (MCNP) transport code. The MCNP transport code was adapted for the specific configurations related to pulsed neutron well logging. The MCNP model probabilistically predicts neutron lifetimes, interactions with surrounding elements, and induced gamma rays, thereby allowing for the forward modeling of specific responses from pulsed neutron well logging instruments, particularly with regard to energy and time spectra.

The pulsed neutron well logging parameters used to obtain Fig. 1 are summarized in Table 1.

Table 1—MCNP modeling parameters

Openhole size	12.25 inches
Casing size	None
Pulsed neutron tool type	Next-generation pulsed neutron tool
Fluid in the hole	1.05 g/cc Water
Lithology	Sandstone
Porosity and fluid	40 pu and 1.05 g/cc water in the pore

The induced gamma rays emitted by neutron-nucleus interactions vary according to the type of incident neutron. High-energy neutrons induce gamma emission through inelastic reactions, whereas thermal neutrons predominantly produce gamma rays via capture reactions. Fig. 1 was obtained through the use of fast, epithermal, and thermal neutrons. Peaks in gamma-ray count rates are identified for principal abundant elements, such as the oxygen peak at 6.13 MeV, along with two escape peaks at 5.6 MeV and 5.1 MeV, indicating the presence of fast neutrons within the measurement medium. Furthermore, the 2.23 MeV peak, which pertains to hydrogen, corroborates the existence of thermal neutrons. Furthermore, the 1.78 MeV silicon peak, along with several energy peaks for chlorine at 1.96 MeV, 1.17 MeV, and 788 kilo-electron volts (keV), as well as the 0.499 keV boron peak.

Spectral Deconvolution

When an energy spectrum is available, predicting lithium concentration involves estimating ^{10}B and ^{35}Cl concentrations through analysis of the energy spectrum. To quantify the ^{10}B and ^{35}Cl content within the brine, their respective spectral signatures are isolated from the spectrum. Standard methods are employed to deconvolve the individual components that constitute an energy spectrum. Such methods include spectral stripping, spectral window approaches, and spectral deconvolution techniques that use spectral elemental standards.

Chlorine Gamma-Ray Spectra. The spectral deconvolution of ^{35}Cl components can be achieved through elemental standard-based spectral analysis. The typical spectrum of ^{35}Cl gamma rays, based on a pulsed neutron source comprising fast, epithermal, and thermal neutrons, is presented in Fig. 2 in blue. The ^{35}Cl spectrum exhibits numerous gamma emissions and associated peaks. ^{35}Cl gamma rays are distributed over a broader energy range.

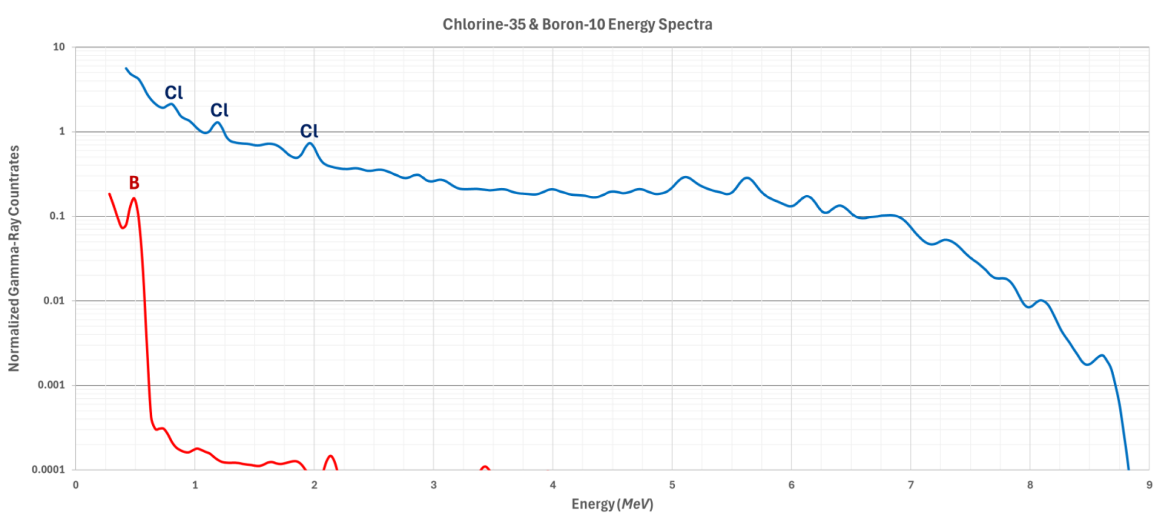

Figure 2—^{35}Cl- and ^{10}B-induced gamma-ray energy spectra from a brine-filled sandstone.

Boron Gamma-Ray Spectrum. ^{10}B engages in (n,α) reactions that result in the emission of gamma rays at an energy of 477 keV. This reaction produces a notably simple gamma energy spectrum, with nearly all gamma rays occurring at the lower end of the energy range, and none exceeding 0.5 MeV. This spectrum is also shown in Fig. 2 in red compared to the ^{35}Cl spectrum. Fig. 3 shows the lower energy section of the spectrum presented in Fig. 2.

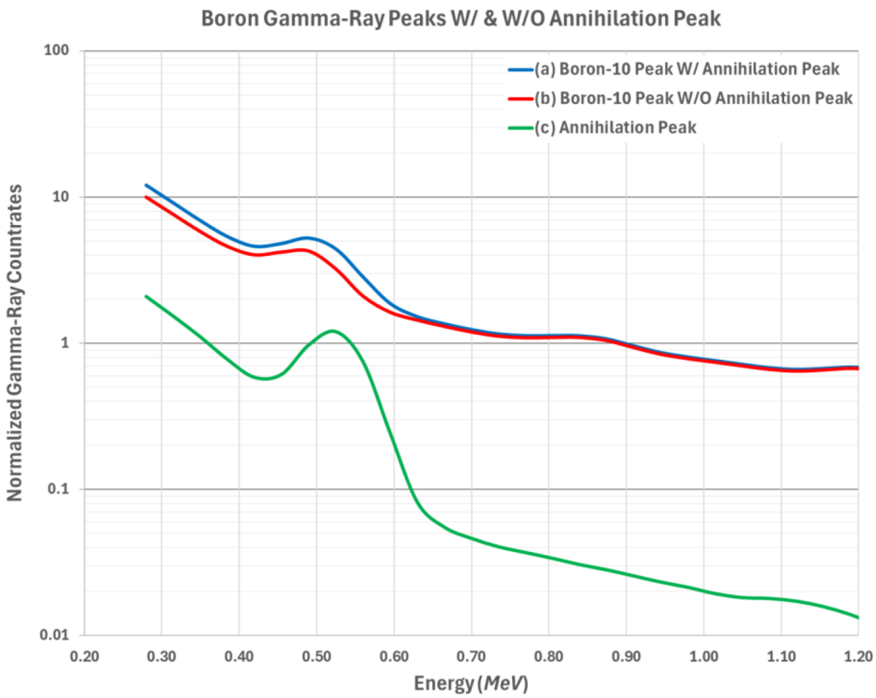

Figure 3—^{10}B-induced gamma-ray peaks (a) including annihilation gamma rays, (b) excluding annihilation gamma rays, and (c) annihilation gamma ray peak from a brine-filled sandstone.

Predicting ^{10}B concentrations is more challenging than predicting ^{35}Cl concentrations. This challenge arises from the presence of high-energy gamma rays in the environment. These high-energy gamma rays induce pair production reactions and lead to the emission of electron-positron pairs. Subsequently, these pairs undergo annihilation, emitting gamma rays with an exact energy of 511 keV (refer to Fig. 3(c) green spectrum). Consequently, the characteristic gamma-ray signature from annihilation often overlaps with the ^{10}B signature, necessitating the removal of the 511 keV annihilation peak from the spectrum to accurately estimate ^{10}B concentration.

Notably, the energy of these annihilation gamma rays closely resembles that of gamma rays emitted by ^{10}B isotopes, as demonstrated in the spectra presented in Fig. 3(a) and (c), where the peaks corresponding to ^{10}B and annihilation are observed to overlap. The ^{10}B peak with the annihilation peak in blue necessitates the removal of the annihilation peak. Once the annihilation peak is eliminated, the identification of the ^{10}B component becomes feasible, as indicated in red (Fig. 3(b)).

Thereafter, the yields of ^{35}Cl and ^{10}B are converted into concentrations using correlations based on experimental data.

Correlating Elemental Concentrations to Time Spectrum

Fundamental Background. The pulsed neutron source's burst and decay periods are controllable, unlike a chemical source-based neutron tool. This feature allows the time spectrum to include gamma rays measured during the source's active phase, which produces high-energy neutrons. Later, during the source's inactive phase, neutrons are slowed down to thermal energies, resulting in gamma ray emissions characteristic of thermal neutron capture reactions.

After the concentrations of ^{10}B and ^{35}Cl are determined from the energy spectra, those elements' concentrations are correlated with time-spectrum-based parameters (or pulsed neutron attributes that can be extracted as logs from the time spectrum). A ratio of gamma ray count rates from the time domain, while predominantly the source is on, including a short period of source off, and those from when the source is off (denoted as the Total-to-Capture ratio) can be obtained. The Total indicates the gamma rays given off via fast, epithermal, and thermal neutrons. Additionally, if the capture gamma rays are excluded from the total gamma ray count rates, this is known as the inelastic counts, and another gamma-ray ratio parameter can be obtained (i.e., Inelastic-to-Capture ratio). The Inelastic-to-Capture ratio is similar to the Total-to-Capture ratio. In this study, we focus on the Total-to-Capture ratio in ^{6}Li concentration delineation.

Stripping Proxy Elements' Responses. A target formation's Total-to-Capture ratio is a combination of individual elements' Total-to-Capture ratios. From the pulsed neutron logging tool measurement, which is the combined Total-to-Capture ratios, each element's Total-to-Capture ratio is derived via the stripping method.

Fig. 4 presents the Total-to-Capture ratio across three distinct borehole and formation scenarios. The initial case involves (a) a borehole and formation containing ^{6}Li and ^{10}B at varying concentrations. The second scenario pertains to (b) a borehole and formation with varying ^{10}B concentrations. Lastly, scenario (c) depicts a borehole and formation fluids containing only ^{6}Li at different concentrations. The horizontal axis represents the concentration of the individual element in ppm, and the vertical axis shows the Total-to-Capture ratio. Since ^{10}B exhibits a larger neutron-alpha (n,α) reaction cross section than ^{35}Cl, it exhibits a greater magnitude of the Total-to-Capture ratio compared to ^{35}Cl. The influence of ^{35}Cl is considerably less significant; therefore, it was excluded from the plot in Fig. 4.

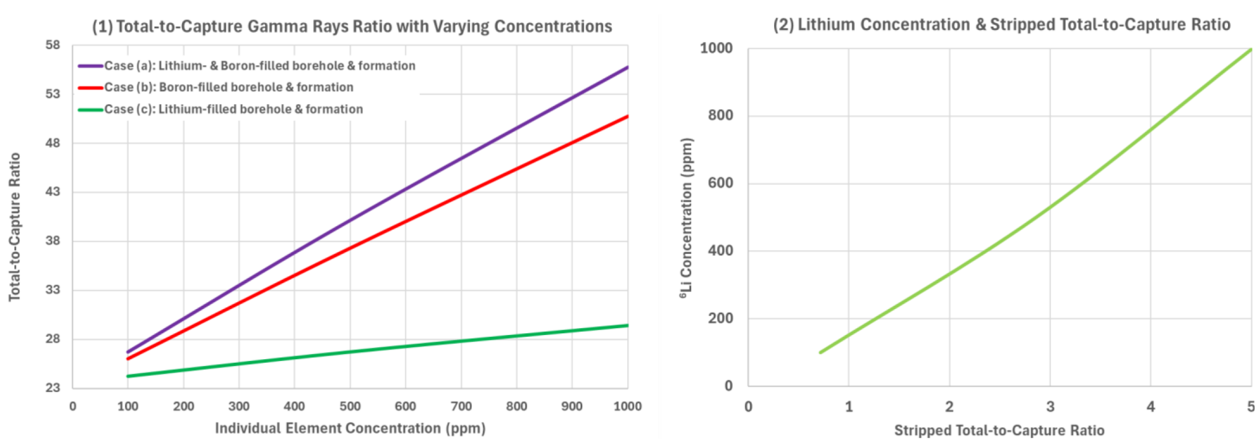

Figure 4—Responses of the Total-to-Capture ratios: (1) (a) a case when ^{6}Li and ^{10}B coexist, (b) a case when ^{10}B is present, and (c) a case when ^{6}Li is present; and (2) a stripped Total-to-Capture ratio concerning varying ^{6}Li concentrations.

Referring to Fig. 4 (1), if the Total-to-Capture ratio exhibits a linear relationship with elements within the borehole and formation, the removal of ^{10}B (**case (b)**) from the presence of ^{6}Li and ^{10}B (**case (a)**) corresponds to that for ^{6}Li (**case (c)**). However, the Total-to-Capture ratio does not demonstrate a linear correlation concerning the concentrations of ^{6}Li and ^{10}B.

Fig. 4 (2) illustrates the stripped Total-to-Capture ratio as a function of variations in ^{6}Li concentrations. Compared to Fig. 4 (1)(c), the observation suggests that the resultant Total-to-Capture ratio does not exhibit cumulative behavior at higher concentration levels. Therefore, a straightforward linear stripping methodology appears insufficient. It may be necessary to conduct further characterization of the Total-to-Capture ratio stripping algorithm for isolate ^{6}Li or to adopt an approach involving model-based interpretation.

This method of stripping pulsed neutron responses for proxy elements can be performed using other parameters, such as the Inelastic-to-Capture ratio or pulsed neutron logs that are obtained from different time gates.

Alternative Approach Using MCNP Simulation. An alternative method, which circumvents the need for stripping pulsed neutron responses for assessing ^{6}Li concentration, employs Monte Carlo simulation-based response charts. After calculating the ^{10}B concentration from the energy spectrum, the Total-to-Capture ratio response charts as presented in Fig. 5 are built for ^{10}B concentrations of (a) 0 ppm, (b) 100 ppm, (c) 500 ppm, and (d) 1,000 ppm, respectively. If necessary, the step sizes in the ^{10}B concentration can be reduced. The horizontal and vertical axes depict porosity and the Total-to-Capture ratio responses, respectively, for all plots.

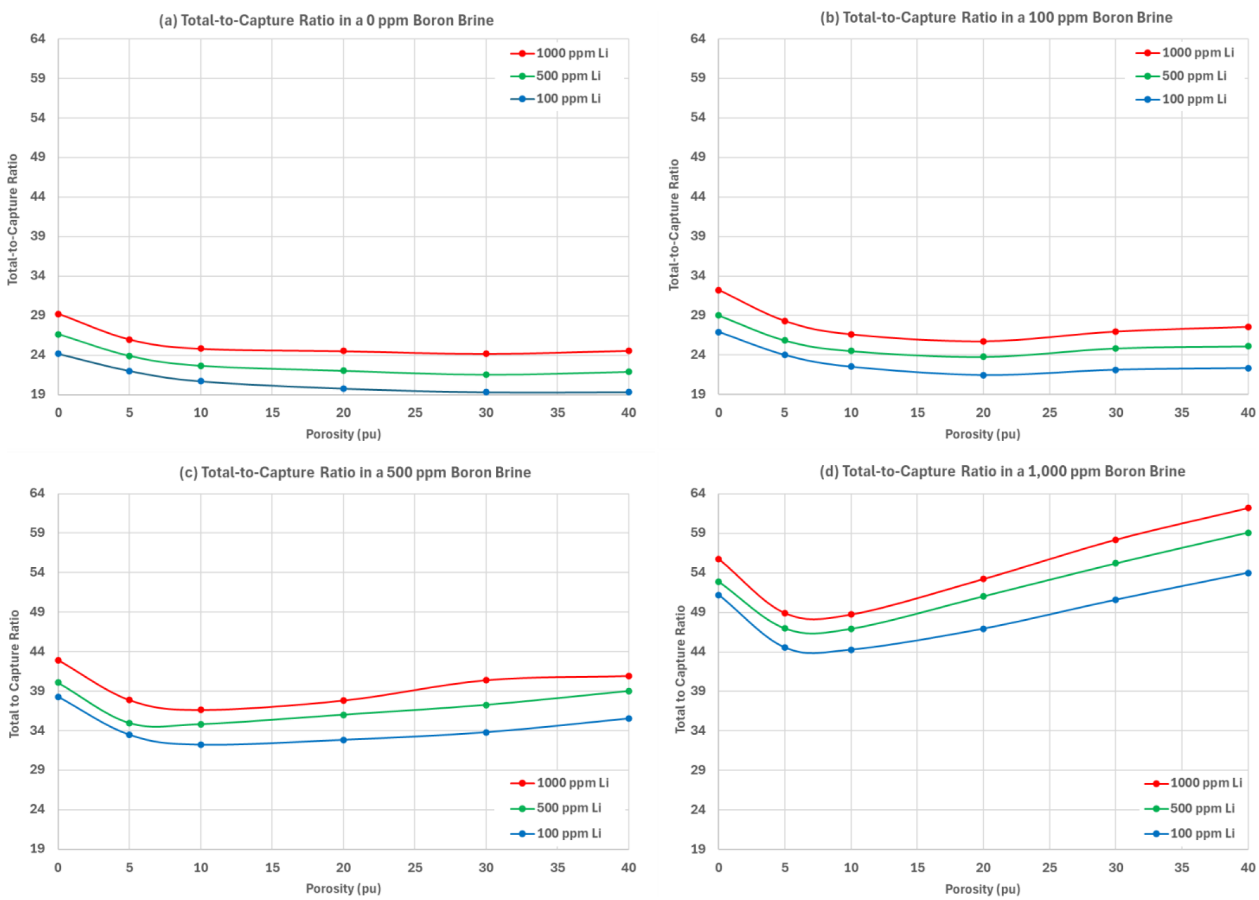

Figure 5—Responses of the Total-to-Capture ratio with respect to ^{10}B concentration, porosity, and ^{6}Li concentration.

The intersection point of the porosity value and the Total-to-Capture ratio at a specific ^{10}B concentration chart provides the ^{6}Li concentration. For example, using the 100 ppm ^{10}B chart from Fig. 5(b) at a 30 pu point, it can be observed that the lithium concentration is 100 ppm if the Total-to-Capture ratio is measured at 22. If the Total-to-Capture ratio is 27, this indicates a brine with a lithium concentration of 1,000 ppm. The Total-to-Capture ratio rises as the ^{6}Li concentration increases at a constant ^{10}B concentration.

The responses of the Total-to-Capture ratio increase as the ^{10}B concentration in brine increases. Therefore, accurately quantifying ^{10}B concentration from the energy spectrum is essential.

From all response charts, the responses from the matrix point (i.e., zero pu) to the five pu formation diminish. The magnitude of decrease is greater with the increase in ^{10}B concentration. As porosity increases from 5 pu to 40 pu, the responses of the Total-to-Capture ratio do not reveal wide variation in each ^{6}Li concentration except for the formation filled with 1,000 ppm ^{10}B brine. A greater dynamic range in the Total-to-Capture ratio shows the correlation with ^{10}B concentration.

The technique detailed for the Total-to-Capture ratio can be applied to other time-spectra-based pulsed neutron logs, such as Inelastic-to-Capture.

Flow Chart for Lithium Concentration Log

Fig. 6 summarizes the procedures for delineating the concentration of ^{6}Li as a continuous well log in a flowchart format.

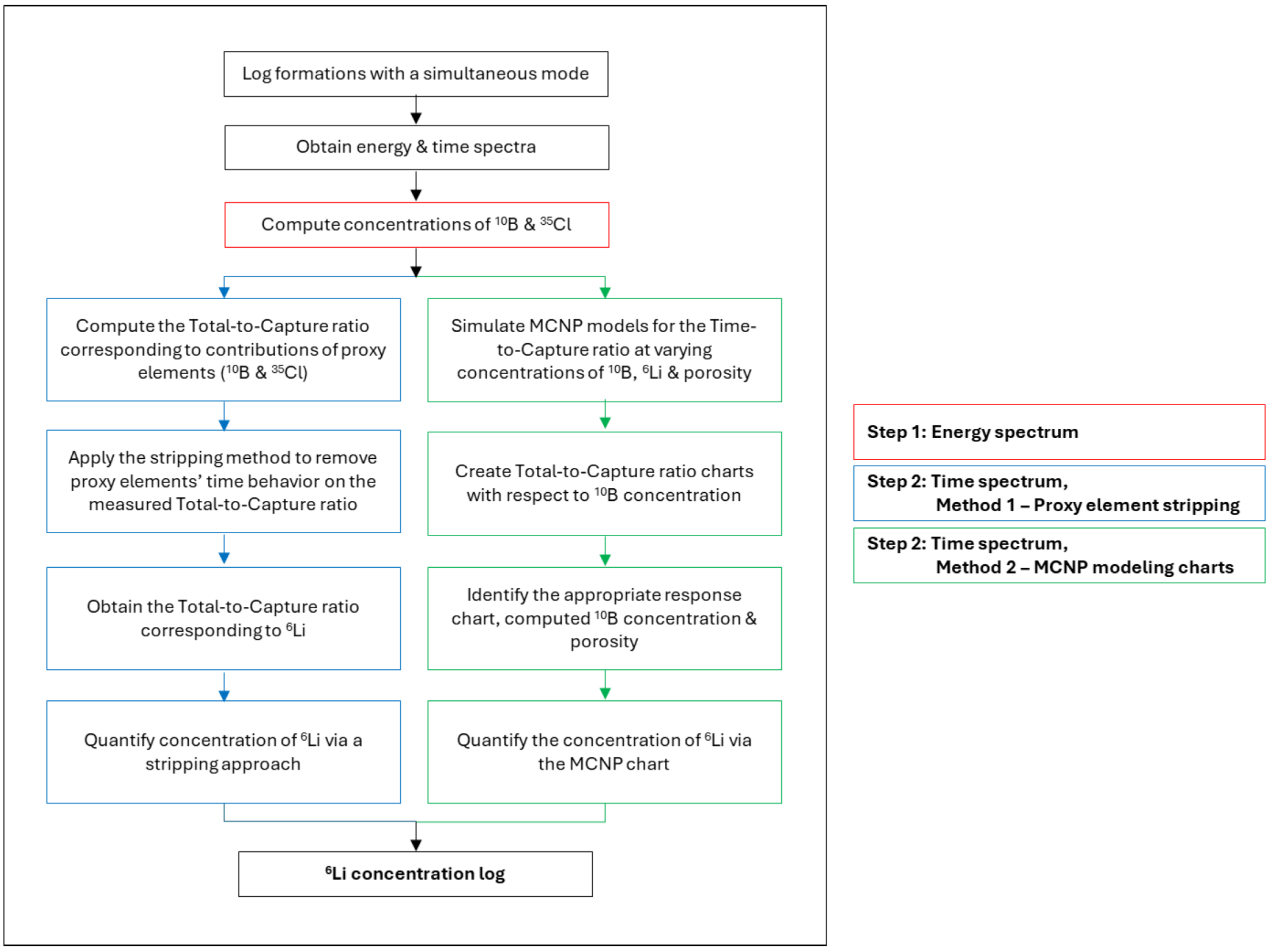

Figure 6—The flow chart illustrating a method from pulsed neutron well logging to the ^{6}Li concentration log: the step within a red box utilizes the energy spectrum (Step 1), the steps within blue boxes employ the time spectrum and a proxy element stripping algorithm (Step 2, Method 1), and those within green boxes make use of the time spectrum and MCNP modeling charts (Step 2, Method 2).

As illustrated on the right side of the flow diagram, the procedure consists of two primary steps (i.e., Step 1 and Step 2). The initial step involves the use of the energy spectrum recorded from a short-spaced detector (denoted by **a red box**). The second step can be categorized into two distinct approaches: proxy element stripping (indicated by **blue boxes**) and MCNP modeling charts (indicated by **green boxes**).

Key Benefits of Using Pulsed Neutron Logging

The advantages of employing pulsed neutron logging technology include:

1. Rapid data acquisition of time and energy spectra in open- and cased-hole completed wells with any well deviation is feasible.
2. Controllability of neutron output enables stable time spectra for both total and capture time domains.
3. ^{6}Li assessment can be conducted on a continuous depth basis rather than a discrete depth basis.
4. The elemental yield deconvolution method and the isolation of proxy elements' concentrations in time spectra can be efficiently streamlined as a hybrid analysis approach.
5. A customized stochastic MCNP modeling is adopted in the data analysis, allowing for characterizing ^{6}Li concentration in diverse scenarios.

Summary and Conclusions

We present a novel hybrid method for assessing lithium concentration levels in a brine-saturated formation through the pulsed neutron well logging technique. The main discussions and conclusions are summarized as follows:

- A two-stage pulsed neutron tool-based measurement system is developed to predict ^{6}Li concentration.
- The first stage is based on the energy spectrum measurements to determine the concentrations of ^{10}B, ^{35}Cl, and any other high thermal neutron capture cross section elements.
- The final prediction of ^{6}Li concentration can be done in two ways.
    - Obtain pulsed neutron logs from the time spectrum, such as the Total-to-Capture ratio and the Inelastic-to-Capture ratio. Subsequently, apply a proxy element stripping algorithm to eliminate the concentrations of ^{10}B and ^{35}Cl calculated from the energy spectrum.
    - Another approach is to form porosity vs. time spectrum parameter charts for various ^{10}B concentrations, where the time spectrum parameters can be the Total-to-Capture ratio and the Inelastic-to-Capture ratio. The charts are formed through Monte Carlo stochastic simulations. Then, use the correct chart by referring to the ^{10}B concentration to determine the ^{6}Li concentration.

Coring, FT&S, and other conventional and advanced well logging data play a key role in building a lithium concentration model and database. ^{6}Li concentration assessment can be conducted using prior knowledge of the formation and environment, such as the correlation between proxy elements and lithium levels.

However, our proposed method does not depend on any pre-existing databases. Instead, it uses energy spectrum and parameters that indicate the temporal behavior of the gamma-ray spectrum, such as the Total-to-Capture ratio and the Inelastic-to-Capture ratio. We either remove the behaviors of proxy elements from the measured time spectrum or employ a simultaneous solution through Monte Carlo simulation-based charts designed for different ^{10}B levels. The contribution of ^{35}Cl is anticipated to be negligible due to its comparatively low thermal neutron capture cross section, allowing it to be effectively managed through background subtraction techniques. Therefore, this method is well-suited for ^{6}Li assessment during the exploration phase when prior knowledge and databases are not readily available.

Acknowledgment

The authors thank Baker Hughes for permission to publish this paper.

References

Akagbosu, P., Kim, Y., Nardiello, R., Chace, D., McGlynn, I., Lawal, K., Akomeno, K., Toumelin, C., and Atuanya, C. 2023. Carbon/Oxygen Logging Speed Optimization in the Niger Delta Fields With Improved Pulsed Neutron

Technology. Paper presented at the SPE Nigeria Annual International Conference and Exhibition, Lagos, Nigeria, 31 July-2 August. SPE-217241-MS.

Chace, D., Trcka, D., and Dawe, B. 1994. Application and Interpretation of Continuous Oxygen Activation Logs for Measuring Complex Water Flow Profiles in Injection Wells. Paper presented at the 69th Annual Technical Conference and Exhibition, New Orleans, Louisiana, USA, 25-28 September. SPE 28412-MS.

Conner, A., Chace, D., Abou-Saleh, J., Kim, Y., Kelley, M., Place, M., Pardini, R., and Gupta, N. 2018. Pulsed Neutron Capture Logging: Monitoring CO_2 Breakthrough in Multiple CO_2-EOR Field. Paper presented at the 14th Greenhouse Gas Control Technologies (GHGT) Conference, Melbourne, Australia, 21-25 October.

Gilchrist, A., Prati, E., and Pemper, R. 1999. Introduction of New Through-Tubing Multifunction Pulsed Neutron Instrument. Presented at the 1999 SPE Annual Technical Conference and Exhibition, Houston, Texas, USA, 3-6 October. SPE-56803-MS.

Inanc, F., Gilchrist, W., and Chace, D. 2009. Physical Basis Modeling, and Interpretation of a New Gas Saturation Measurement for Cased Wells. Paper presented at the SPWLA 50th Annual Logging Symposium, Woodlands, Texas, USA, 21-24 June. SPWLA-2009-42298.

Kim, Y., Akagbosu, P., Ndokwu, C., and Adesokan, I. 2024a. Three-Phase Saturation Evaluation Utilizing Multidetector Pulsed Neutron Logging in Freshwater Environments: A Guide for Sidetrack Drilling Optimization. Paper presented at the SPE Nigeria Annual International Conference and Exhibition (NAICE), Lagos, Nigeria, 5-7 August. SPE-221716-MS.

Kim, Y., Al-Quoud, K., Sahib, M., and Sergeev, E. 2023c. Post-Waterflooding Oil Reservoir Surveillance Using Pulsed Neutron Well Logging in Upper Burgan Formation, Northern Kuwait. Presented at the Gas & Oil Technology Showcase and Conference held in Dubai, UAE, 13-15 March. SPE-214263-MS.

Kim, Y., and Abdel Aziz, A. 2022. Delineation of Multi-Phase Wellbore and Formation Fluid Distribution in Abu Dhabi's Onshore Horizontal Wells. Paper presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, 31 Oct-3 Nov 2022. SPE-211647-MS.

Kim, Y., and Akang, N. 2024b. Crucial Factors for In-situ CO_2 Monitoring in CO_2-EOR Projects Using Pulsed Neutron Well Logging. Paper presented at the 2024 Gas & Oil Technology Showcase and Conference (GOTECH), Dubai, UAE, 7-9 May. SPE-219169-MS.

Kim, Y., and Chace, D. 2016. Simultaneous Solution of Three-Phase Formation Fluid Saturations Using Triangulation of Multidetector Pulsed Neutron Measurements. Paper presented at the International Petroleum Technology Conference, Bangkok, Thailand, 14-16 November. IPTC-18687-MS.

Kim, Y., Boyle, K., Chace, D., Akagbosu, P., Oyegwa, A., Wyatt, D., Okowi, V., and Gade, S. 2017. Innovative Triangulation Technique Enables Simultaneous Determination of Three-Phase Formation Fluid Saturations Using Pulsed Neutron Logging. Paper presented at SPWLA 58th Annual Logging Symposium, Oklahoma City, Oklahoma, USA, 17-21 June. SPWLA-2017-GG.

Kim, Y., Gray, R., and Odusi, G. 2023d. Pressure Depletion Delineation For A Deep Gas Condensate-producing Well In The North Sea. Paper presented at the 2023 SPE Annual Technical Conference and Exhibition, San Antonio, Texas, USA, 16-18 October. SPE-214826-MS.

Kim, Y., Kotov, A., and Chace, D. 2021. Time-Lapse Pulsed Neutron Well Logging in Oil Sands for Monitoring Steam Chamber Development. Paper presented at the 2021 SPE Middle East Oil & Gas Show and Conference, Manama, Bahrain, 28 November-1 December. SPE-204576-MS.

Kim, Y., Kotov, A., Chace, D., Yuan, P., Anniyev, T., Inanc, F., and McGlynn, I. 2023a. Efficient Through-Casing Surveillance of Steam Chamber and Reservoir Oil Using a New Pulsed Neutron Technology and an Advanced Interpretation Algorithm. Paper presented at the 2023 SPE Middle East Oil, Gas and Geosciences Show, Manama, Bahrain, 19-21 February. SPE-213317-MS.

Kim, Y., Lim, L., and Chace, D. 2025a. Distinguishing CO_2 and Hydrocarbon Gas Using Pulsed Neutron Logs for CO_2 Storage Projects in Depleted Gas Reservoirs. Paper presented at the SPWLA 66th Annual Logging Symposium, Dubai, UAE, 17-21 May. SPWLA-2025-0083.

Kim, Y., Odusi, G., Edwards, W., and Hooker, A. 2025b. Time-lapse Monitoring of Pressure Depletion for High-Temperature Gas-Condensate Reservoirs to Optimize In-Fill Perforation Using Pulsed-Neutron Well Logging. Paper presented at the Middle East Oil, Gas and Geosciences Show, Manama, Bahrain, 16-18 September. SPE-227464-MS.

Kim, Y., Puri, A., Sahu, S., and Tyagil, A. 2023b. Accurate Identification of Gas-Bearing Formation in a Mature Field Using Pulsed Neutron Logs Prevented Well Abandonment. Presented at the Gas & Oil Technology Showcase and Conference held in Dubai, UAE, 13-15 March. SPE-214153-MS.

McGlynn, I., Anniyev, T., Inanc, F., Chace, D., Kotov, A., Yuan, P., Soans, E., and Batubara, A. 2023. Development and Baseline Comparison of a New Pulsed-Neutron Spectroscopy Tool for Carbon-Oxygen Analysis and Three-Phase Saturation Monitoring. Paper presented at the SPWLA 64th Annual Logging Symposium, Lake Conroe, Texas, USA, 10-14 June. SPWLA-2023-0007.

Nardiello, R., Chace, D., McGlynn, I., Akagbosu, P., Ouadah, S, Ramaswamy, V., Toumelin, E., Oduwole, A., Ogofa, M., and Ekeng, M. 2022. New Development In Pulsed Neutron Technologies To Enhance Faster Data Recovery And Reduce Operation Logging Time For Three-Phase Saturation Analysis. Paper presented at SPWLA 63rd Annual Logging Symposium, Stavanger, Norway, 10-15 June. SPWLA-2022-0020.

Nardiello, R., Kim, Y., Chace, D., Zhang, Q., Galli, G., Pirrone, M., Borghi, M. 2017. Cased Hole Reservoir Pressure Analysis Using Pulsed Neutron Log Measurements in Challenging Mature Environments: Physics, Modeling, Uncertainty Assessment and Application. Paper presented at the SPWLA 58th Annual Logging Symposium held in Oklahoma City, Oklahoma, USA, June 17-21. SPWLA-2017-TTT.

Randall, R. 1985. Application of accelerator sources for pulsed neutron logging of oil and gas wells. *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms, Volumes 10-11*, Part 2, Pages 1028–1032.

Trcka, D., Berg, F. V., Peeters, M., and Mickael, M. 1996. Measuring Three-Phase Holdups in Horizontal Wellbores Using Pulsed Neutron Instruments. Paper presented at the 1996 SPE Annual Technical Conference and Exhibition, Denver, Colorado, USA, 6-9 October. SPE-36561-MS.

Trcka, D., Gilchrist, A., Riley, S., Bruner, M., Esfandiari, T., Ly, T., Shearin, D., Patino, T., Murray, H., Potter, J., Rose, R., Chen, J., Olsen, S., Lovera, O., McCants, D., Berger, A., Ellsworth, K., Barolak, G., Martain, R., McFall, A., and Guo, P. 2006. Field Trials of a New Method for the Measurement of Formation Gas Using Pulsed-Neutron Instrumentation. Paper presented at the 2006 SPE Annual SPE Technical Conference and Exhibition, San Antonio, Texas, USA, 24-27 September. SPE-102350-MS.

Youmans, A., Hopkinson, E., and Steward, R. 1959. (D,T) Neutron Activation Logging. Paper presented at the Fall Meeting of the Society of Petroleum Engineers of AIME, Dallas, Texas, USA, 3-6 October. SPE-1304-G.

Youmans, A., Millis, W, Hopkinson, E., and Bishop, D. 1964. The Neutron Lifetime Log. Paper presented at the SPWLA 5th Annual Logging Symposium, Midland, Texas, USA, 13-15 May. SPWLA-1964-M.